WSQ COURSE SLIDES

AI Agents with Gemini Spark

Agent strategy · vibe coding · human review · governed automation



Course Code: TGS-2026064173

Version v1.0 · 23 August 2026

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome and Operating Boundaries

Everything needed before the technical learning begins

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Digital Attendance (Mandatory)

1

AM attendance

Scan the trainer-provided TRAQOM QR code when class begins.

2

PM attendance

Complete the second attendance after lunch.

3

Assessment attendance

Complete the assessment attendance before the PP begins.

4

Funding control

At least 75% attendance and competency in all assessments are required.

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About the Trainer



Your Trainer

General Trainer template

NAME

DESIGNATION

QUALIFICATIONS

EXPERTISE

EXPERIENCE

CONTACT

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About the Trainer



Dr. Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Principal trainer and courseware architect
QUALIFICATIONS	Doctorate; ACTA/DACE-qualified adult educator
EXPERTISE	AI agents, cloud automation, software engineering and digital learning
DELIVERS	Technical WSQ programmes and applied workplace labs
ORGANISATION	Tertiary Infotech Academy Pte Ltd
CONTACT	enquiry@tertiaryinfotech.com

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ICEBREAKER

Learner Introductions

1

Your role

What work do you own today?

2

Your repetitive task

Which workflow consumes time or attention?

3

Your web goal

What should the website help a visitor accomplish?

4

Your risk boundary

Which action should always need human approval?

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Ground Rules

1

Participate

Ask questions and share evidence, not credentials.

2

Protect data

Use mock data and provided accounts only.

3

Work in small changes

Review every meaningful patch before continuing.

4

Keep control

Stop an agent that exceeds the approved plan.

5

Return on time

Breaks are short in this one-day programme.

6

Respect ownership

Do not publish, send or transact without approval.

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FOUR TECHNICAL BLOCKS

Course Outline

1

Agent Strategy and Gemini Spark Runtime

LO1 · K1, A1

2

Vibe Coding a Responsive Website

LO2 · K4, K5, A2

3

Skills, Schedules and Implementation Governance

LO3 · K3, K5, A3

4

Monitoring, Incident Control and Optimisation

LO4 · K2, K6, A4

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SCHEDULE

One-Day Lesson Plan

Time	Learning block	Evidence
09:20-11:20	Agent strategy + Lab 1	Strategy and approval map
11:20-13:40	Vibe coding + Lab 2	Responsive website and test log
13:40-15:30	Skills/schedules + Lab 3	Skill and implementation plan
15:30-16:30	Monitoring + Lab 4	Scorecard and incident record
16:40-18:00	PP + OQ	A1-A4 and K1-K6 evidence

DECISION RULE Detailed procedures remain in the Learner Guide and lab folders.

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WHAT YOU WILL BE ABLE TO DO

Learning Outcomes

1

LO1

Determine an AI-agent strategy by translating a web-design goal into bounded tasks, permissions, risks and human approval points.

2

LO2

Develop web content and a responsive website through a controlled vibe-coding loop using Claude Code, Codex or Antigravity, with Gemini Spark supporting research and reusable skills.

3

LO3

Develop an implementation plan for Gemini Spark tasks, skills, schedules and connected tools, including responsibilities, timelines and rollback controls.

4

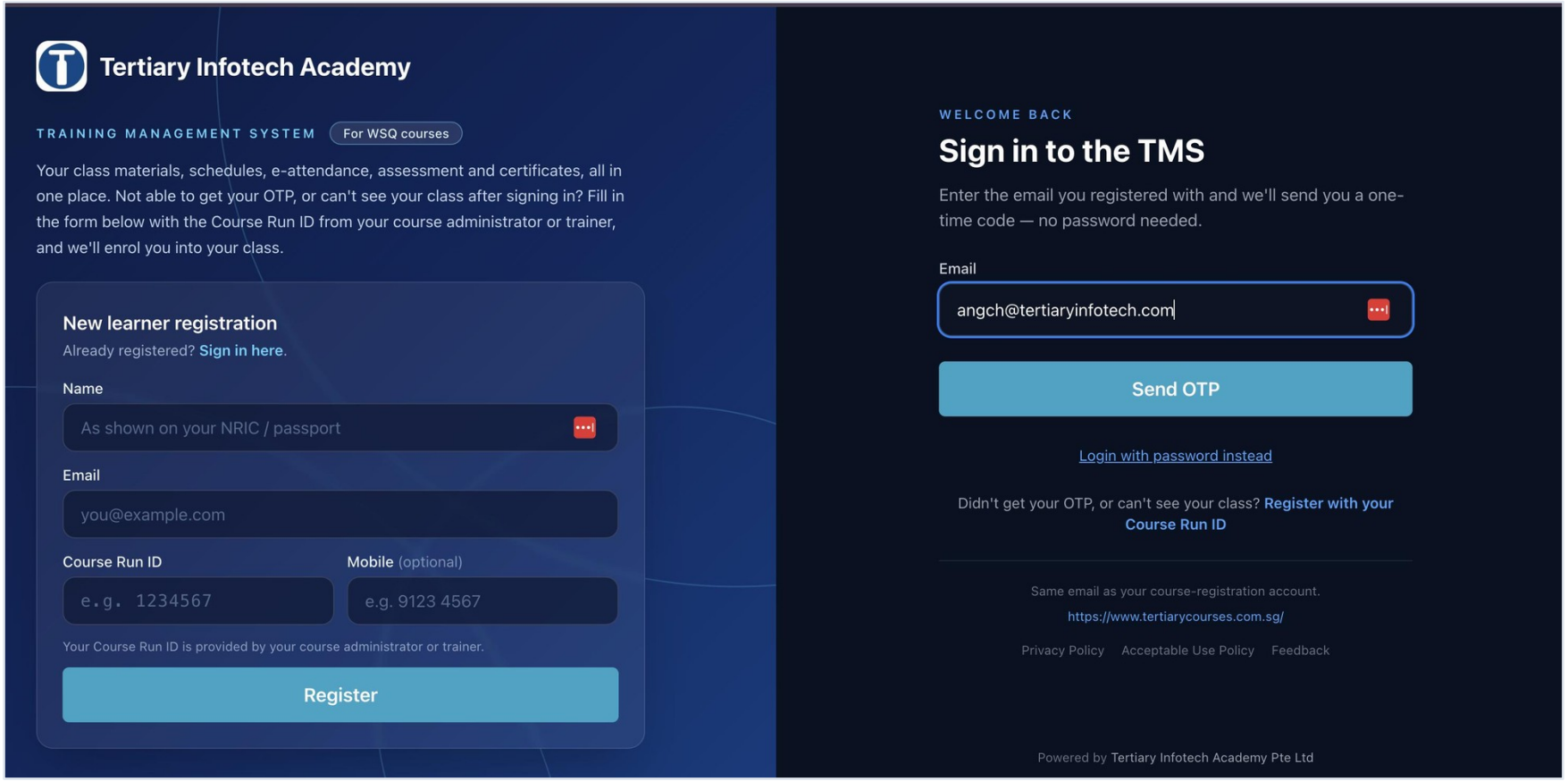
LO4

Evaluate agent and website performance using evidence, acceptance criteria, incident signals and iterative improvement decisions.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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Download Course Materials

1

LMS course record

The verified course links are stored at lms-tms.tertiaryinfotech.com.

2

Learner Guide

Use the detailed procedures, prompts, evidence rules and acceptance checks.

3

Learner slides

Use the PDF for concepts, visual models and assessment alignment.

https://lms-tms.tertiaryinfotech.com/

4

Hands-on labs

Get the self-contained lab folders from the linked GitHub repository.

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Access the Hands-On Labs

1

Option A · Clone

git clone https://github.com/tertiarycourses/TGS-2026064173-AI-Agents-with-Gemini-Spark.git

2

Option B · Download

GitHub → Code → Download ZIP

3

Lab structure

Each lab is self-contained with files, evidence and acceptance checks.

https://github.com/tertiarycourses/TGS-2026064173-AI-Agents-with-Gemini-Spark

4

Coding choice

Use Claude Code, Codex or Antigravity; record which one you chose.

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ASSESSMENT RULES

Briefing for Assessment

1

Individual

Complete the PP and OQ independently.

2

Open book

Use the slides, Learner Guide and approved lab files.

3

Evidence

Submit the requested artifacts and answer each criterion.

4

Competency

Every A and K criterion must be assessed competent.

5

No sharing

Do not photograph or circulate assessment scripts.

6

Appeal

The assessor will explain the feedback and appeal process.

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Assessment Design

1

Practical Performance

4 tasks · A1-A4 · 60 minutes

2

Oral Questioning

4 questions · K1-K6 · 20 minutes

3

Submission

Candidate papers are completed on the LMS.

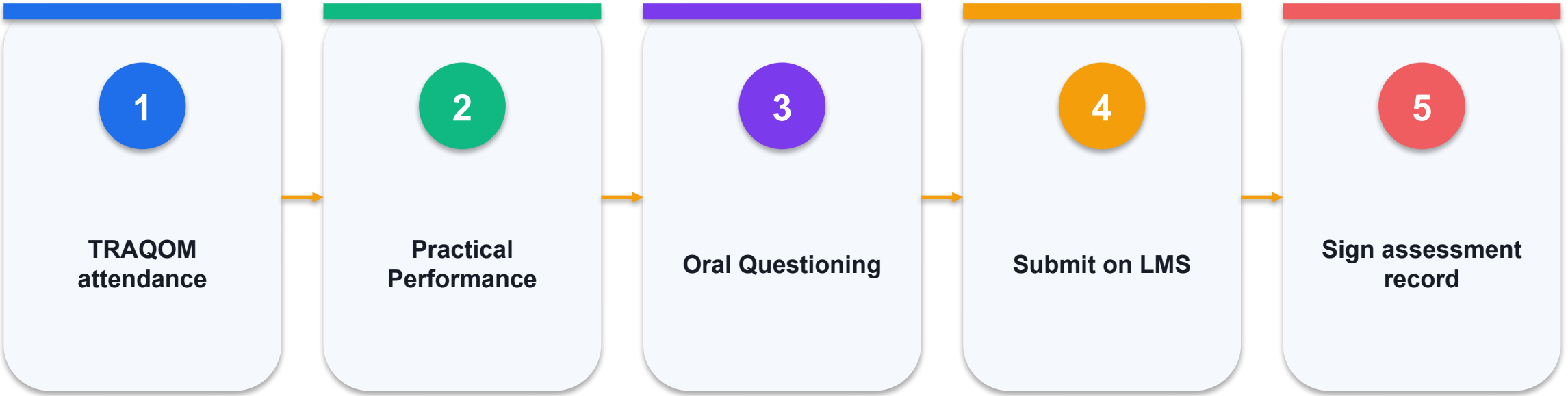
4

Privacy

Answer keys remain trainer-only and are never published.

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Assessment Flow



THE POINT PP covers A1-A4; OQ covers K1-K6; all criteria must be competent.

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CORE MECHANISMS

From Chatbot to Governed Agent

The technical distinctions that make the rest of the course work

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RUNTIME MODEL

Chatbot, Assistant and Agent

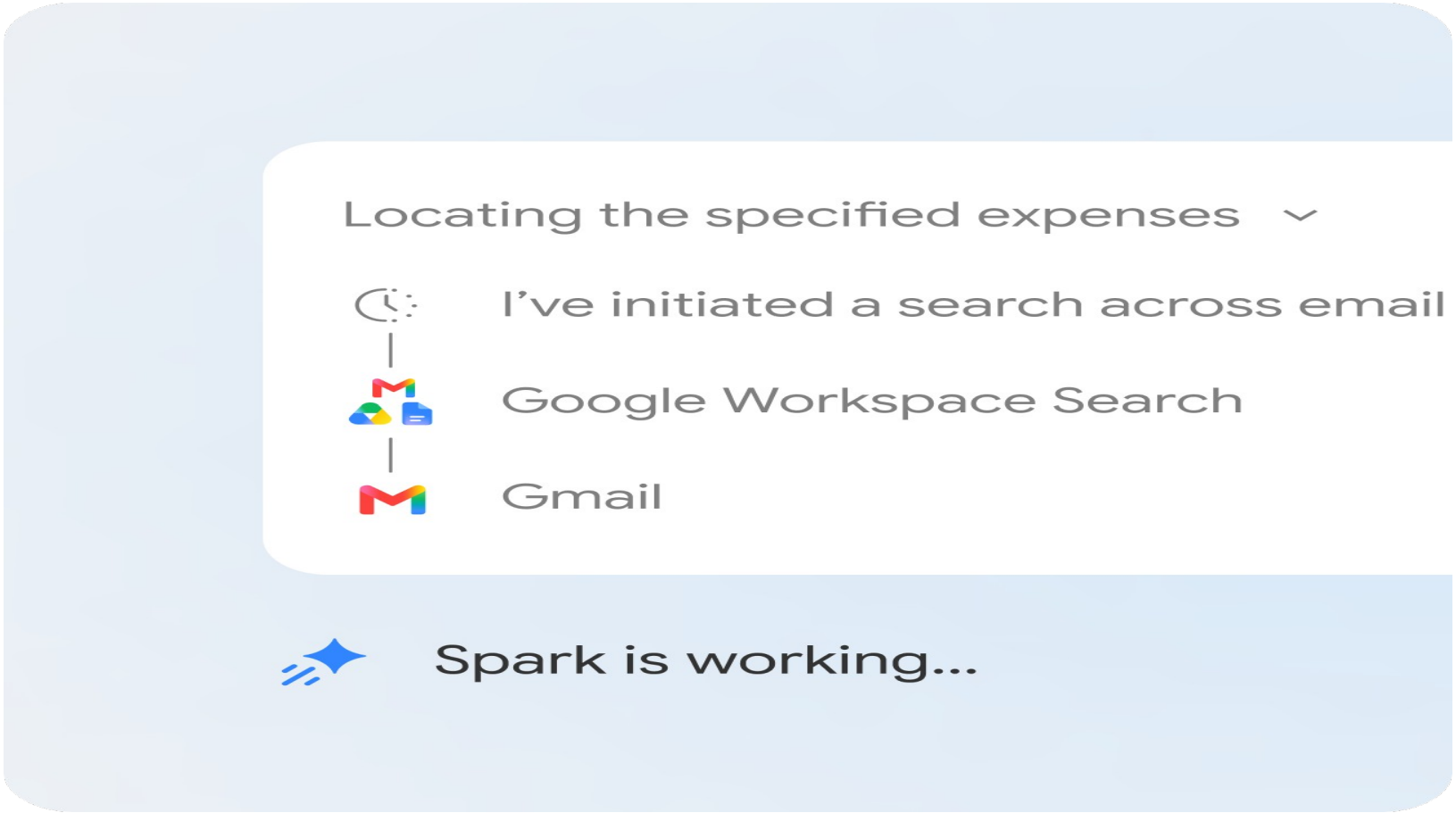
Property	Chatbot	Agent
Control loop	Prompt → response	Goal → plan → act → observe → revise
State	Conversation context	Task thread, files, schedules and tool results
Tools	Optional retrieval	Connected apps, browser and code execution
Time	Ends with response	May continue in background or on a trigger
Risk	Wrong answer	Wrong action or repeated wrong action

DECISION RULE Treat every tool action as a state change that needs evidence and a recovery route.

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OFFICIAL PRODUCT EVIDENCE

Gemini Spark: Tasks, Skills and Schedules

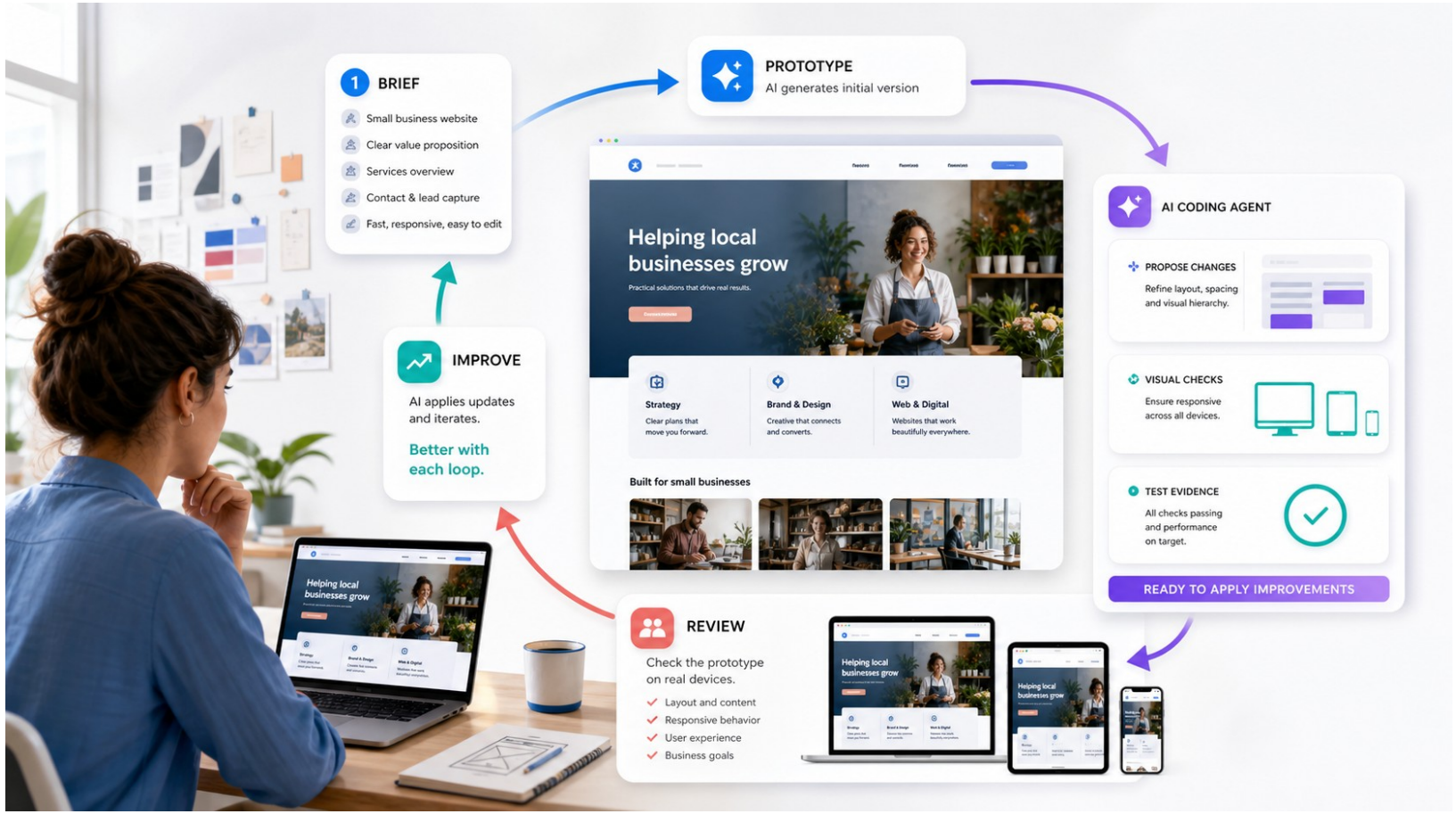


READ THE VISUAL

Tasks define what, skills define how, and schedules define when. Connected apps provide the action surface; the task thread and work panel preserve observable progress.

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Source ↗

The Vibe-Coding Loop

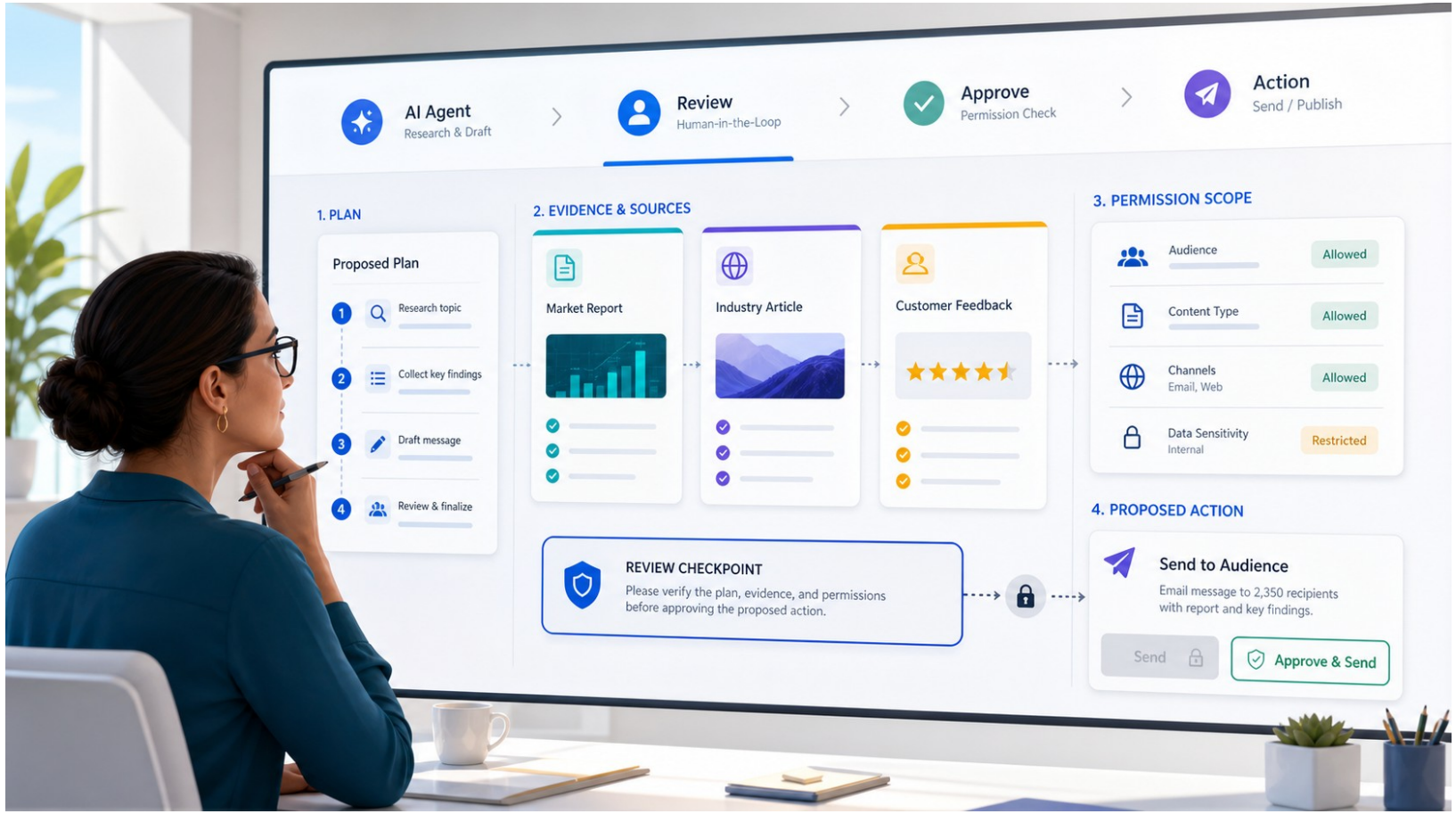


READ THE VISUAL

A useful loop is brief → thin prototype → diff review → browser checks → owner review → one controlled improvement. The human owns intent and acceptance.

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Human-in-the-Loop Is a Control System



READ THE VISUAL

The reviewer checks plan, evidence, permissions and consequence before the final action. A decorative approve button is not enough; the evidence and rollback route must be visible.

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CURRENT LIMITS

Product Constraints That Change the Design

Constraint	Current product evidence	Design consequence
Account	Personal Google Account; Pro or Ultra; Keep Activity on	Prepare a fallback simulation for learners without access
Availability	Feature availability varies by region and platform	Do not make course completion depend on one live screen
Concurrency	Up to 15 tasks may run at once	Use deduplication and no-overlap schedules
Sensitive data	Do not type sign-in or payment data into the task thread	Use takeover and enter it on the website directly
Safety	Experimental feature; supervision remains important	Require review gates and safe stop conditions

DECISION RULE Verify live access before class; use the supplied offline artifacts when Spark is unavailable.

Google Help source ↗

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01

TOPIC 01 · LO1 · K1, A1

Agent Strategy and Gemini Spark Runtime

Goal contract - task plan - tools - state - approval - evidence

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DESIGN PRINCIPLE

Agent Strategy and Gemini Spark Runtime

Autonomy is safe only when the goal, permission boundary, evidence and stop conditions are explicit.

Use this as the design test for every mechanism, task and lab artifact in this topic.

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Architecture and Trust Boundaries

1

Task thread

Holds the goal, instructions, files and task state.

2

Planner

Decomposes the goal into observable steps and revises when blocked.

3

Skills

Reusable how-to instructions and context, selected explicitly or automatically.

4

Connected tools

Workspace apps, websites, local or remote browser, and code execution.

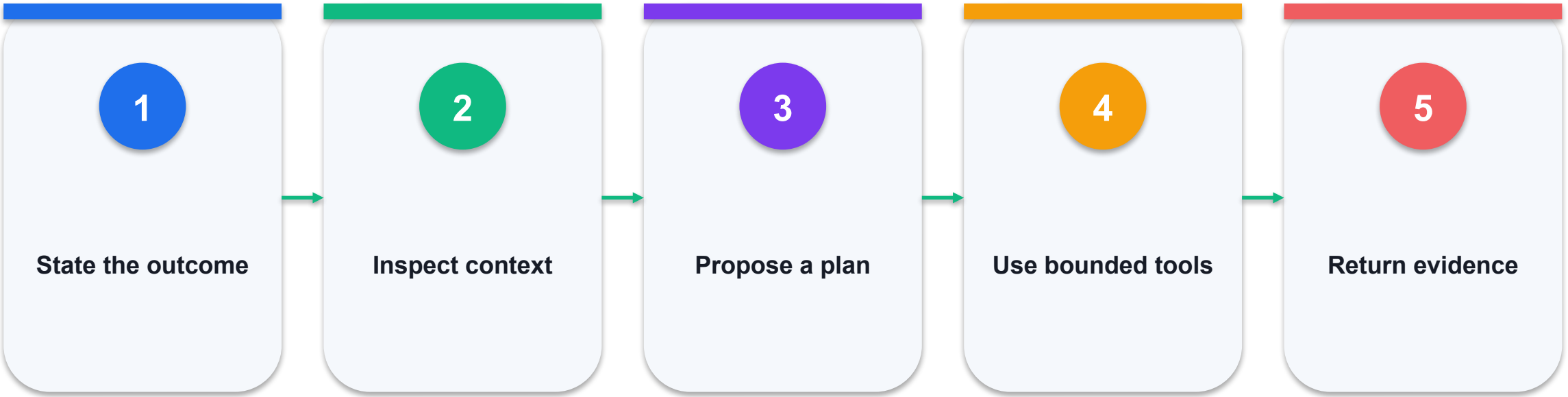
5

Human control

Review, confirm, take over, stop, pause schedules and inspect changed files.

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Source ↗

Runtime Trace



THE POINT A task is observable only when each stage exposes inputs, state, output and the next decision.

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The Concrete Contract

1

Goal

One measurable business outcome; avoid vague verbs such as improve or optimise.

2

Scope

Named apps, folders, sites, audience, date range and excluded data.

3

Output

Exact artifact, schema, destination and quality threshold.

4

Control

Approval gates, stop rules, rollback owner and escalation path.

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Make the Choice Explicit

Property	Option A	Option B
Trigger	User prompt	Time or event
Persistence	Single task thread	Runs repeatedly until paused
Best fit	Bounded deliverable	Stable recurring operation
Primary risk	Misread intent	Unobserved repeated error

DECISION RULE Choose the lowest-autonomy pattern that still meets the business outcome.

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Failure → Diagnostic → Control

Failure	Observable diagnostic	Control / recovery
Goal drift	Plan no longer serves the stated business outcome	Restate acceptance criteria and cancel non-essential branches
Permission creep	Agent requests broader access mid-task	Deny, narrow the task, and document the exception
Hidden dependency	Website, login or file is unavailable	Stop and request takeover; do not improvise credentials
Weak evidence	Agent reports success without artifact or trace	Require links, filenames, before/after state and validation

DECISION RULE Stop on ambiguous authority; do not hide a partial or failed state.

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Worked Case: Northstar Studio

1

Business context

A local design studio needs a lead-generation website. Spark may research competitors and organise source material, while the coding agent builds locally. Publishing, form submission and customer communications remain human-approved actions.

2

Human owner

Approves sources, claims, permissions and any public action.

3

Agent boundary

Researches, drafts, edits or analyses only within the approved contract.

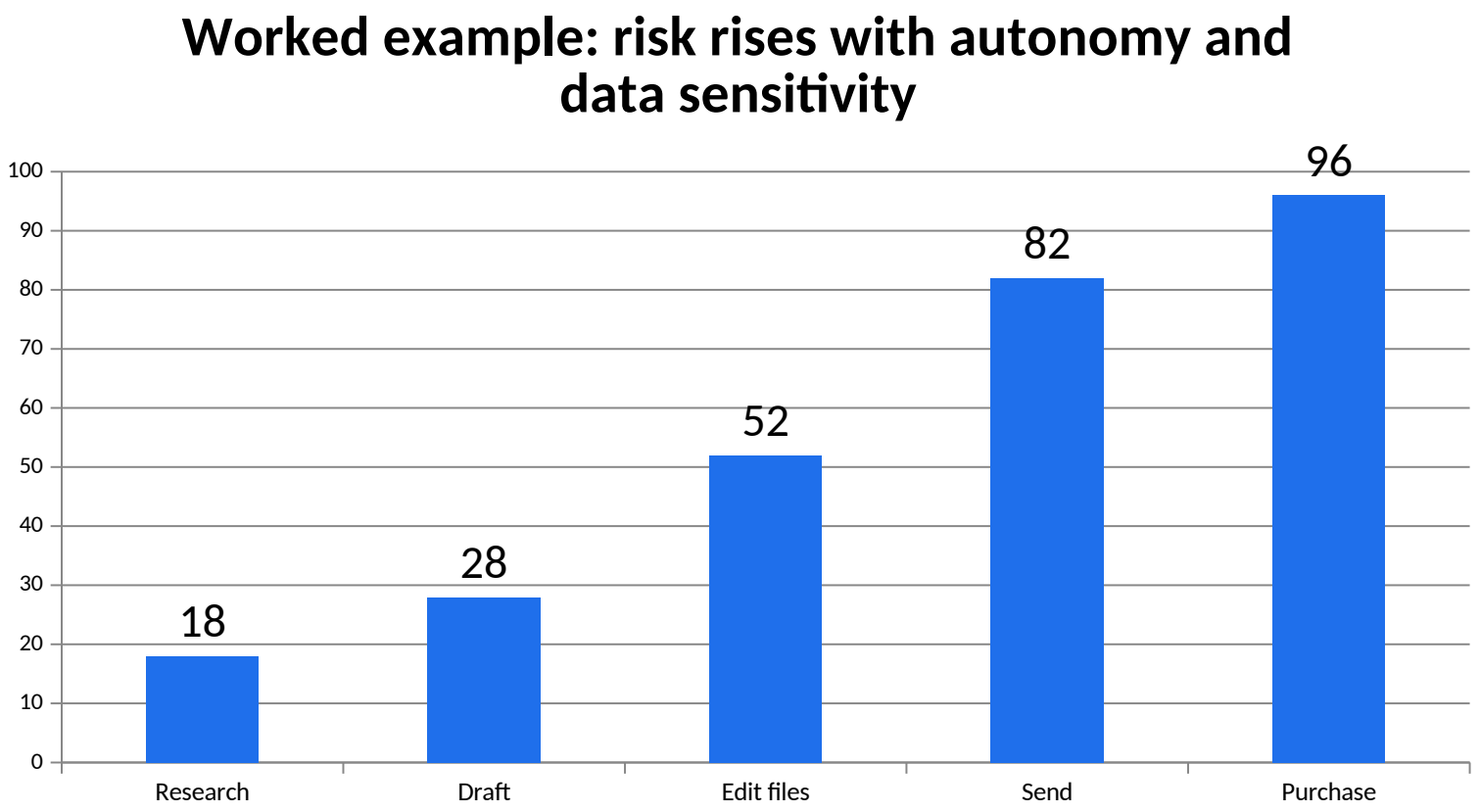
4

Evidence package

Plan, diff, files, tests, source log, approval and rollback point.

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Worked example: risk rises with autonomy and data sensitivity



HOW TO READ IT

1. Values are an illustrative worked example.
2. Compare relative risk or improvement, not industry benchmarks.
3. Use your own run evidence to replace these values.

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Control Points Across the Lifecycle

1

Before

Validate goal, inputs, access and risk tier.

2

During

Observe progress, changed files, tool use and confirmation requests.

3

Before consequence

Require human review for communications, publishing, payments and sensitive edits.

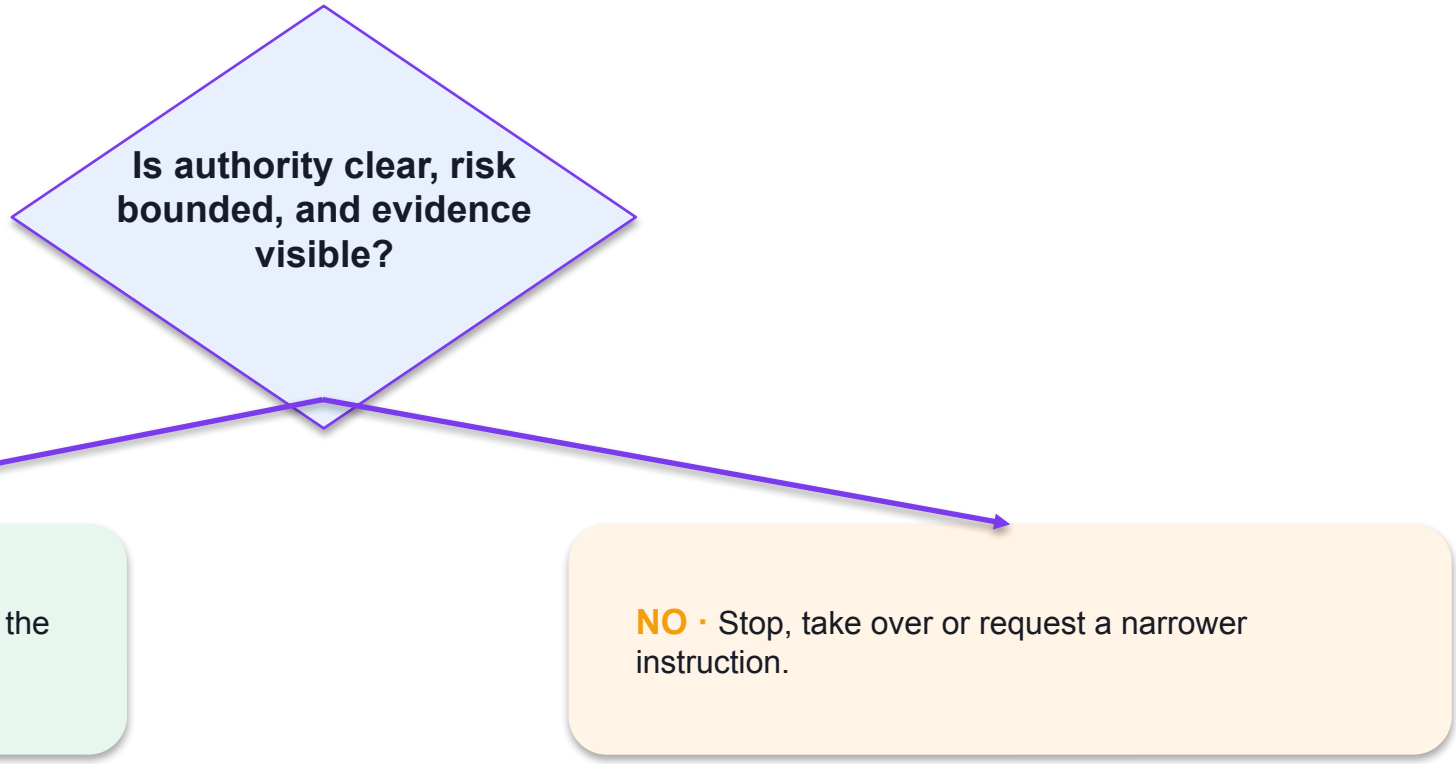
4

After

Read back the external state, store evidence and schedule the next review.

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When May the Agent Continue?



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Evidence Is a Data Model

1

Input evidence

Approved brief, source list, version and permission scope.

2

Process evidence

Plan, task progress, tool calls, changed files and confirmations.

3

Output evidence

Artifact, link, commit, screenshot, test report and source trace.

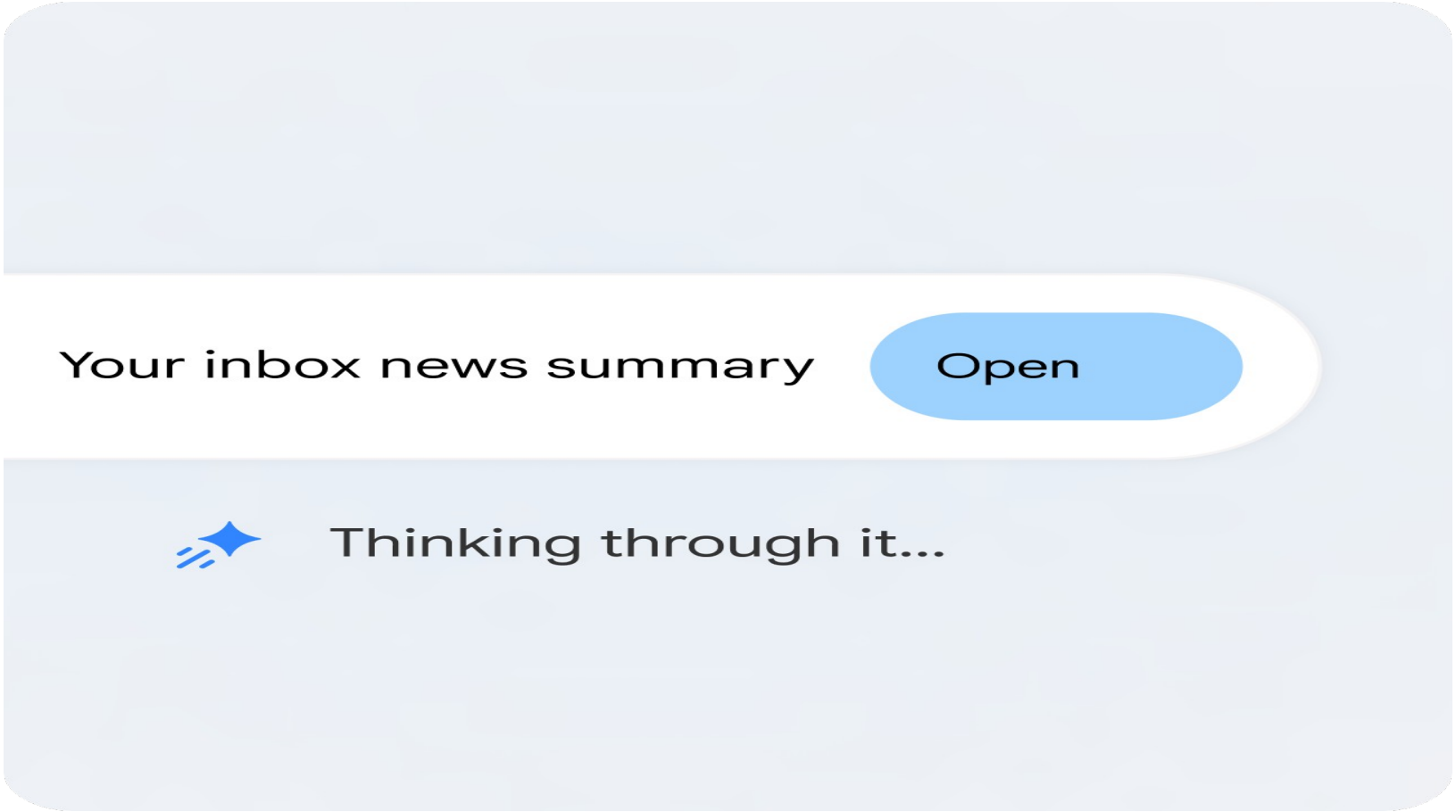
4

Decision evidence

Reviewer, timestamp, approve/reject reason and rollback target.

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Official Spark Example: Inbox Synthesis



READ THE VISUAL

The agent searches a defined information space, synthesises a result and exposes progress. A workplace version still needs source scope, freshness, approval and evidence rules.

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Source ↗

Lab 1 · Create the Agent Strategy and Approval Map

1

Scenario

Translate a website goal into a bounded Gemini Spark task with permissions, risk tiers, approval gates and acceptance evidence.

2

Build

A strategy brief, task contract, permission matrix, risk register and human-approval map for the Northstar Studio website.

3

Tools

Gemini Spark task planning, browser research, Google Drive, human review

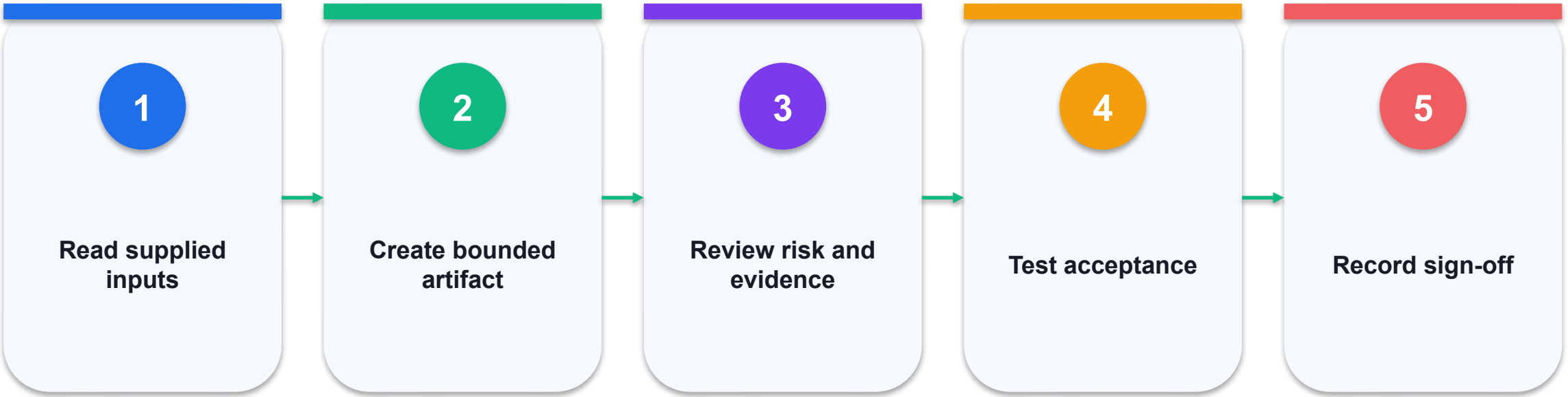
4

Criteria

A1, K1

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Lab Mechanism - Not the Click Path



THE POINT The task can be executed without guessing: every step has an owner, allowed tools, a risk tier, approval rule, stop condition and observable completion evidence.

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Lab Evidence and Acceptance

1

Primary artifact

A strategy brief, task contract, permission matrix, risk register and human-approval map for the Northstar Studio website.

2

Acceptance test

The task can be executed without guessing: every step has an owner, allowed tools, a risk tier, approval rule, stop condition and observable completion evidence.

3

Assessment bridge

Practical criterion: A1

4

Knowledge bridge

Oral criterion: K1

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Topic Technical Takeaways

1

Architecture

Holds the goal, instructions, files and task state.

2

Runtime

State the outcome → Inspect context → Propose a plan → Use bounded tools → Return evidence

3

Control

Restate acceptance criteria and cancel non-essential branches

4

Practice

The task can be executed without guessing: every step has an owner, allowed tools, a risk tier, approval rule, stop condition and observable completion evidence.

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02

TOPIC 02 · LO2 · K4, K5, A2

Vibe Coding a Responsive Website

Brief - prototype - inspect - test - review - improve

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DESIGN PRINCIPLE

Vibe Coding a Responsive Website

Vibe coding is rapid co-creation with an AI coding agent, disciplined by small changes, executable evidence and human review.

Use this as the design test for every mechanism, task and lab artifact in this topic.

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Architecture and Trust Boundaries

1

Design brief

Audience, job-to-be-done, content, constraints and acceptance tests.

2

Coding agent

Claude Code, Codex or Antigravity edits the repository from natural-language intent.

3

Local project

HTML, CSS, JavaScript, assets, README and version history.

4

Test loop

Open in a browser, check responsive layouts, links, form states and accessibility.

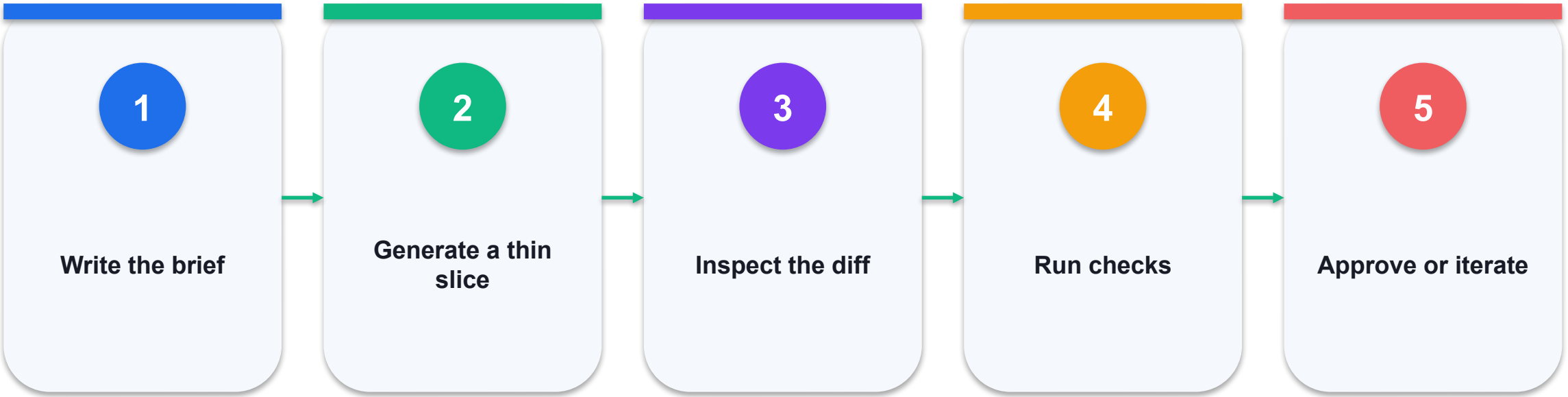
5

Reviewer

Accepts, rejects or redirects each meaningful change before release.

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Runtime Trace



THE POINT A task is observable only when each stage exposes inputs, state, output and the next decision.

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The Concrete Contract

1

Brief

Who the site is for, what action matters, required sections and content source.

2

Guardrails

No secrets, no unapproved dependencies, no destructive commands, preserve working files.

3

Test

Desktop/mobile widths, keyboard navigation, readable contrast, valid links and form feedback.

4

Handoff

README, file map, known limitations, rollback commit and next review date.

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TOPIC 02 · DECISION TABLE

Make the Choice Explicit

Dimension	Choice 1	Choice 2	Choice 3
Interaction	Terminal/repository	Agent manager/IDE	Cloud coding workspace
Typical choice	Claude Code	Antigravity	Codex
Strength	Deep local context	Multi-agent orchestration	Sandboxed agentic coding
Review evidence	Diff and commands	Task artifacts	Patch, tests and run log

DECISION RULE Choose the lowest-autonomy pattern that still meets the business outcome.

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Failure → Diagnostic → Control

Failure	Observable diagnostic	Control / recovery
Prompt pile-up	One huge instruction creates many coupled changes	Ask for one thin vertical slice at a time
Visual regression	Desktop improvement breaks mobile	Test at 1440, 768 and 390 px before approval
Plausible content	Agent invents facts, testimonials or prices	Use a supplied content inventory and source every claim
Dependency sprawl	Unnecessary framework or package added	Require a justification and prefer existing stack

DECISION RULE Stop on ambiguous authority; do not hide a partial or failed state.

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Worked Case: Northstar Studio

1

Business context

Northstar Studio needs a fast, accessible one-page website with a hero, services, proof, and contact section. The team uses a coding agent for the implementation while Spark organises approved source material and revision requests.

2

Human owner

Approves sources, claims, permissions and any public action.

3

Agent boundary

Researches, drafts, edits or analyses only within the approved contract.

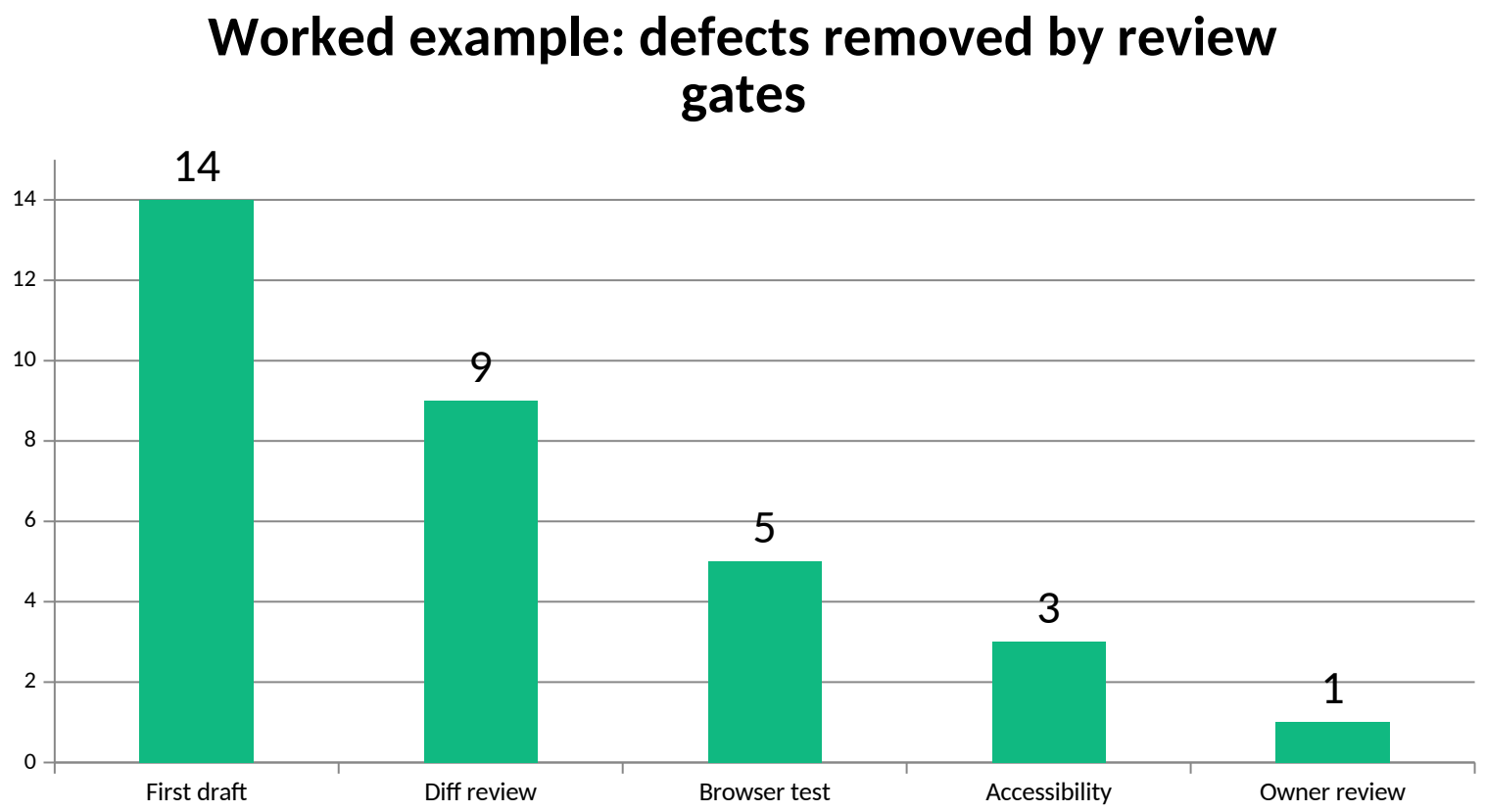
4

Evidence package

Plan, diff, files, tests, source log, approval and rollback point.

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Worked example: defects removed by review gates



HOW TO READ IT

1. Values are an illustrative worked example.
2. Compare relative risk or improvement, not industry benchmarks.
3. Use your own run evidence to replace these values.

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Control Points Across the Lifecycle

1

Before

Validate goal, inputs, access and risk tier.

2

During

Observe progress, changed files, tool use and confirmation requests.

3

Before consequence

Require human review for communications, publishing, payments and sensitive edits.

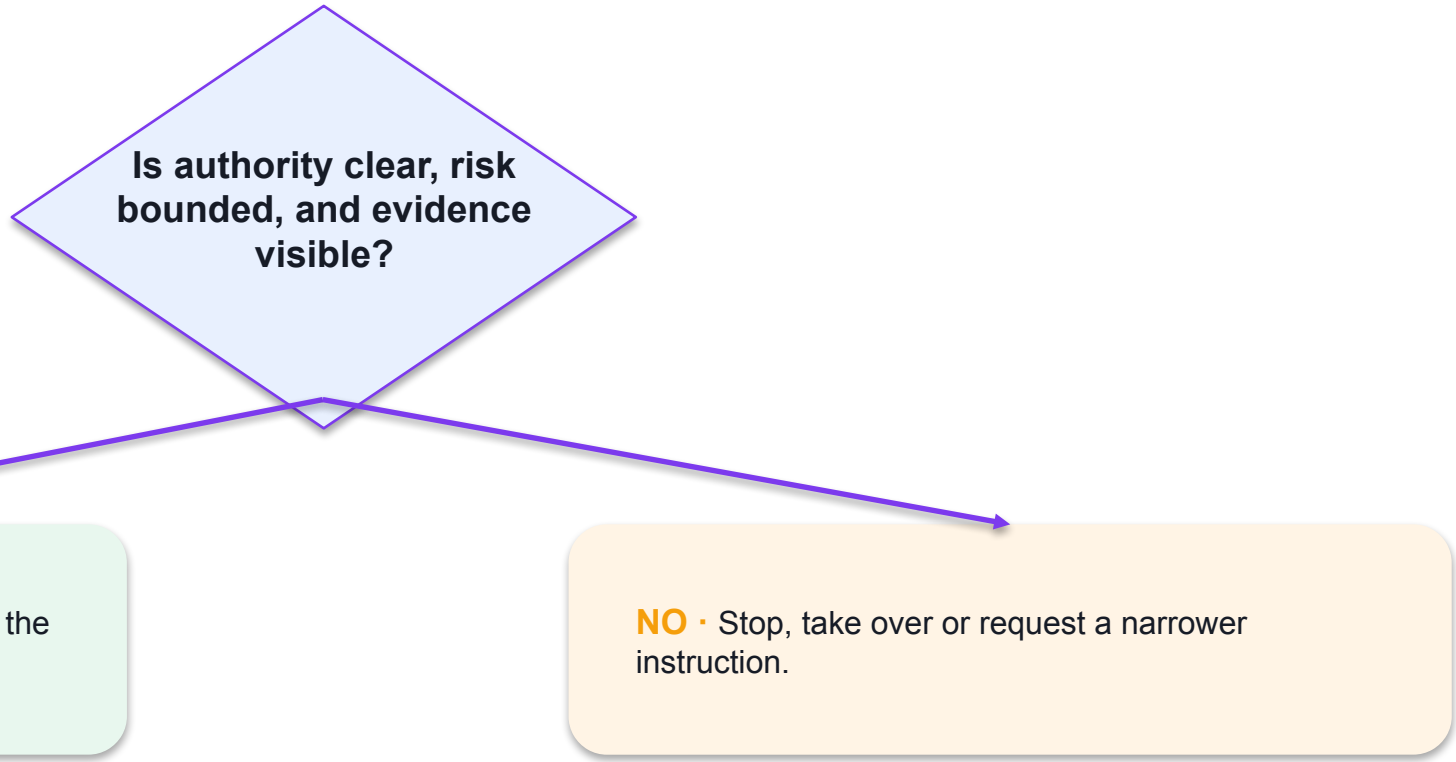
4

After

Read back the external state, store evidence and schedule the next review.

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When May the Agent Continue?



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Evidence Is a Data Model

1

Input evidence

Approved brief, source list, version and permission scope.

2

Process evidence

Plan, task progress, tool calls, changed files and confirmations.

3

Output evidence

Artifact, link, commit, screenshot, test report and source trace.

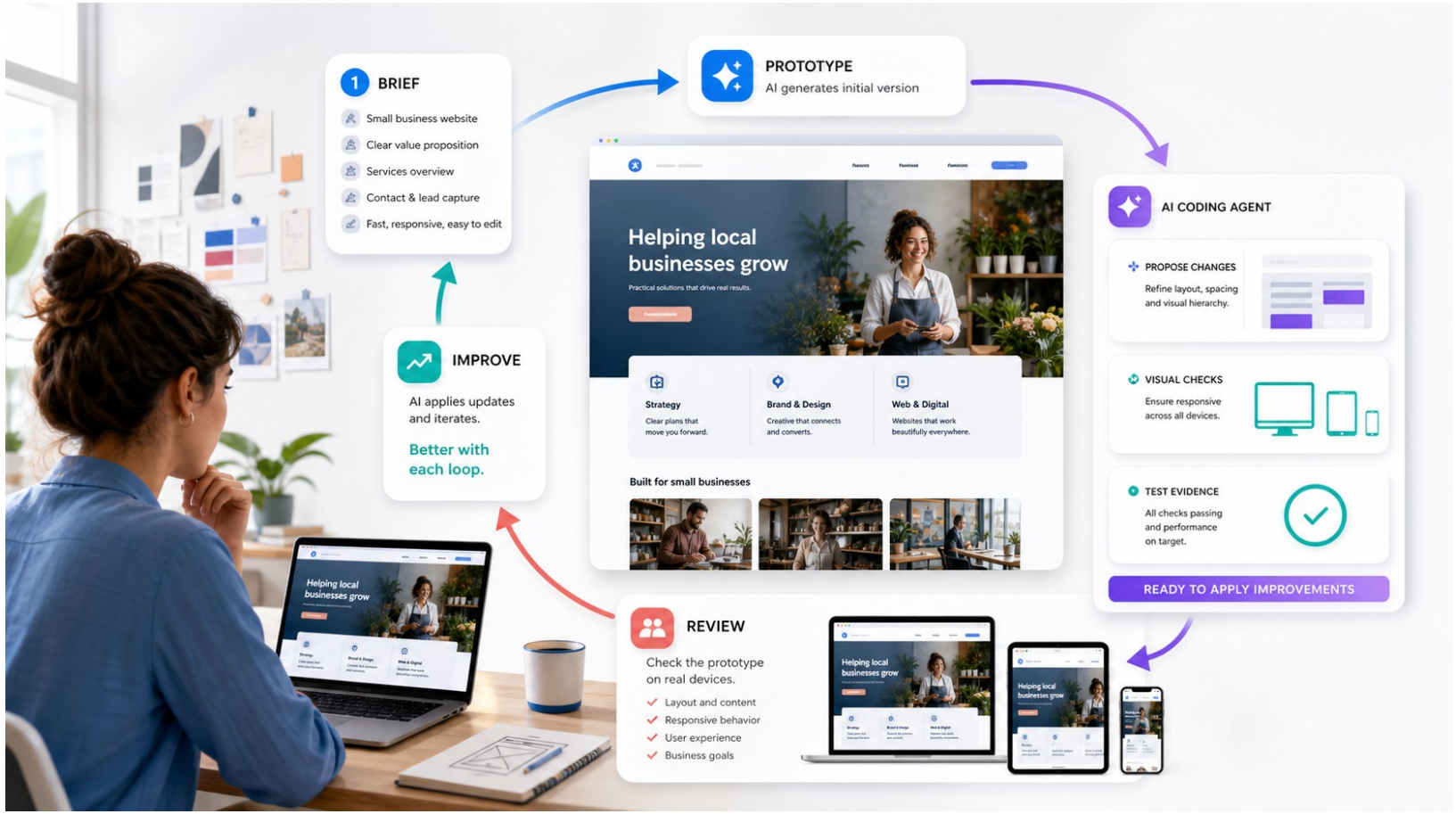
4

Decision evidence

Reviewer, timestamp, approve/reject reason and rollback target.

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Review the Website as an Artifact



READ THE VISUAL

The loop couples repository changes with visual evidence at multiple viewport widths. The coding agent proposes; the human accepts the user experience and business claims.

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Lab 2 · Vibe Code the Northstar Studio Website

1

Scenario

Create and test a responsive website through a bounded vibe-coding loop using Claude Code, Codex or Antigravity.

2

Build

A working one-page HTML/CSS/JavaScript website, responsive at three widths, with accessible navigation, an evidence log and a rollback commit.

3

Tools

Claude Code, Codex or Antigravity; Git; HTML; CSS; JavaScript; browser testing

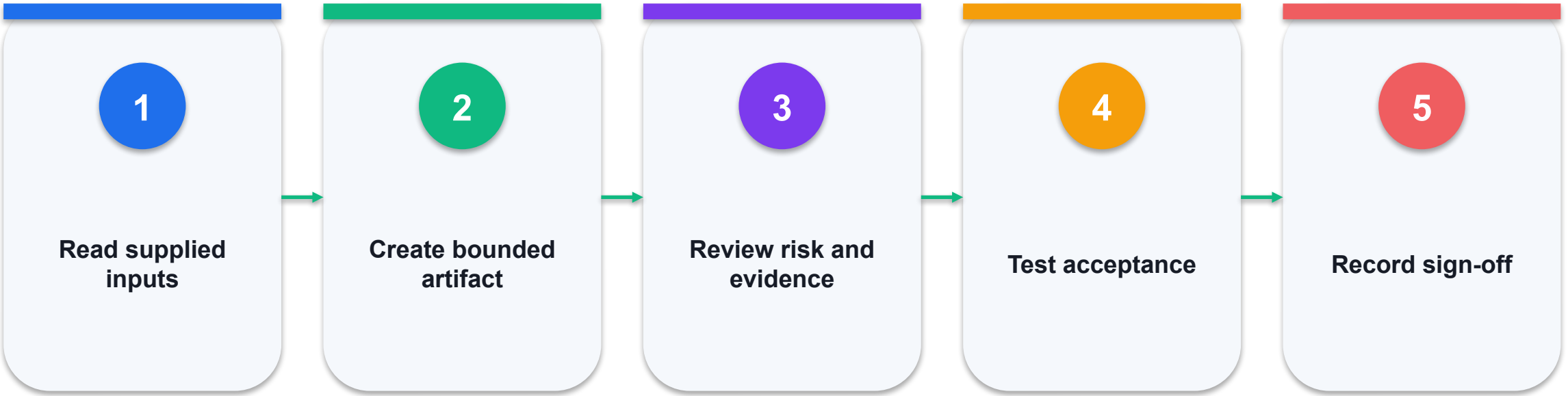
4

Criteria

A2, K4, K5

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Lab Mechanism - Not the Click Path



THE POINT The website renders without errors, passes the supplied structural checks, works at desktop/tablet/mobile widths, is keyboard navigable, contains only approved claims, and has an identifiable rollback commit.

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Lab Evidence and Acceptance

1

Primary artifact

A working one-page HTML/CSS/JavaScript website, responsive at three widths, with accessible navigation, an evidence log and a rollback commit.

2

Acceptance test

The website renders without errors, passes the supplied structural checks, works at desktop/tablet/mobile widths, is keyboard navigable, contains only approved claims, and has an identifiable rollback commit.

3

Assessment bridge

Practical criterion: A2

4

Knowledge bridge

Oral criterion: K4, K5

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Topic Technical Takeaways

1

Architecture

Audience, job-to-be-done, content, constraints and acceptance tests.

2

Runtime

Write the brief → Generate a thin slice → Inspect the diff → Run checks → Approve or iterate

3

Control

Ask for one thin vertical slice at a time

4

Practice

The website renders without errors, passes the supplied structural checks, works at desktop/tablet/mobile widths, is keyboard navigable, contains only approved claims, and has an identifiable rollback commit.

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03

TOPIC 03 · LO3 · K3, K5, A3

Skills, Schedules and Implementation Governance

Reusable instructions - connected apps - trigger - approval - rollback

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DESIGN PRINCIPLE

Skills, Schedules and Implementation Governance

A reusable agent workflow is an operating procedure with versioned instructions, scoped tools, owners and measurable completion evidence.

Use this as the design test for every mechanism, task and lab artifact in this topic.

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Architecture and Trust Boundaries

1

Task

The what: a complete project or objective managed in a thread.

2

Skill

The how: reusable instructions, context, tools and quality rules.

3

Schedule

The when: a time-based or event-based trigger for a task.

4

Connected apps

The where: Gmail, Calendar, Drive, Docs, Sheets, Slides, Search and approved third parties.

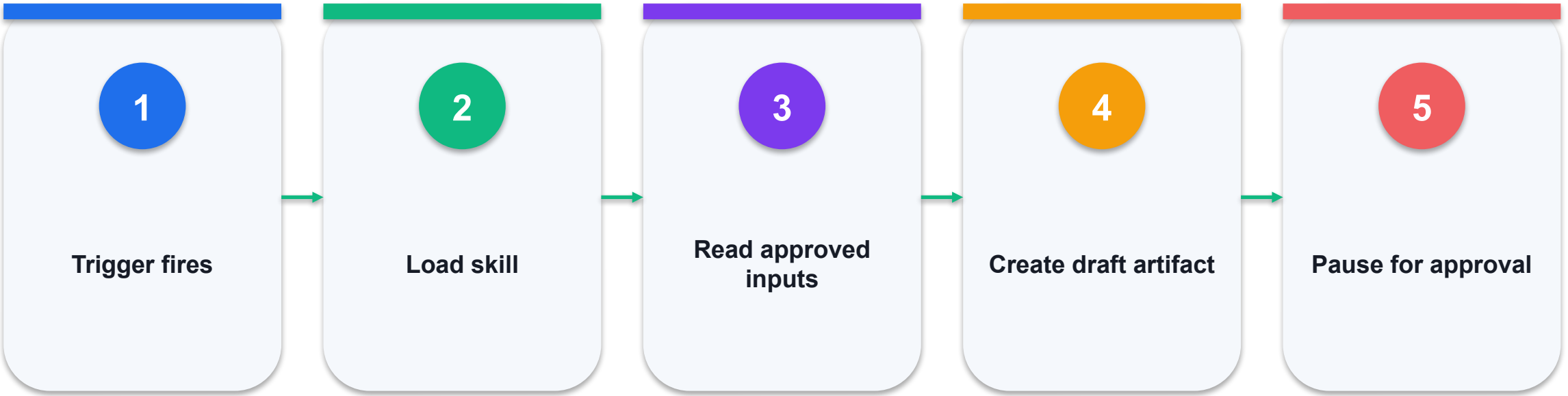
5

Runbook

Owner, approvals, logs, exception handling, rollback and review cadence.

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Runtime Trace



THE POINT A task is observable only when each stage exposes inputs, state, output and the next decision.

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The Concrete Contract

1

Skill trigger

Clear description of when the skill should and should not be used.

2

Inputs

Named folder, file types, required fields and freshness window.

3

Output schema

Required headings, fields, filename and destination.

4

Failure route

Retry limit, alert recipient, safe partial state and manual recovery.

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Make the Choice Explicit

Dimension	Choice 1	Choice 2	Choice 3
One-off	Task	No schedule	Review in thread
Reusable method	Skill	Invoked on demand	Version skill instructions
Recurring	Task + skill	Time/event schedule	Monitor every run
Sensitive	Task + takeover	No unattended run	Human executes final action

DECISION RULE Choose the lowest-autonomy pattern that still meets the business outcome.

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Failure → Diagnostic → Control

Failure	Observable diagnostic	Control / recovery
Stale context	Skill uses an old price list or brief	Validate modification date and reject stale input
Trigger storm	Event creates repeated overlapping runs	Deduplicate by event ID and cap concurrent tasks
Silent partial	Draft created but tracker not updated	Define atomic acceptance evidence for every stage
No owner	Approval request sits unresolved	Assign primary and backup owner with response window

DECISION RULE Stop on ambiguous authority; do not hide a partial or failed state.

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Worked Case: Northstar Studio

1

Business context

A weekly website-content workflow reads an approved content inventory, drafts one update, stores it in a review folder, and notifies the owner. It never publishes automatically; the owner sends an approved change request to the coding agent.

2

Human owner

Approves sources, claims, permissions and any public action.

3

Agent boundary

Researches, drafts, edits or analyses only within the approved contract.

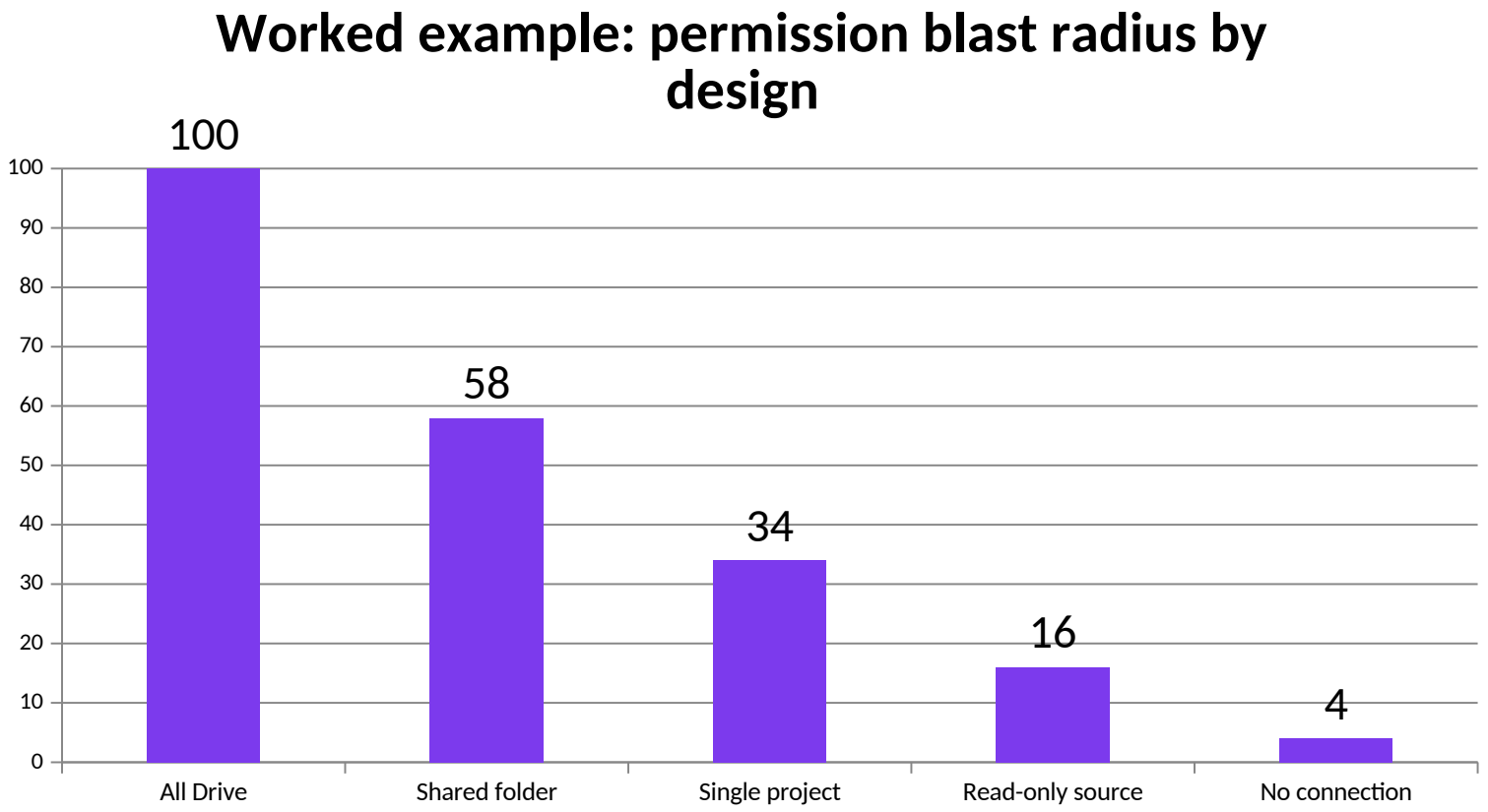
4

Evidence package

Plan, diff, files, tests, source log, approval and rollback point.

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Worked example: permission blast radius by design



HOW TO READ IT

1. Values are an illustrative worked example.
2. Compare relative risk or improvement, not industry benchmarks.
3. Use your own run evidence to replace these values.

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Control Points Across the Lifecycle

1

Before

Validate goal, inputs, access and risk tier.

2

During

Observe progress, changed files, tool use and confirmation requests.

3

Before consequence

Require human review for communications, publishing, payments and sensitive edits.

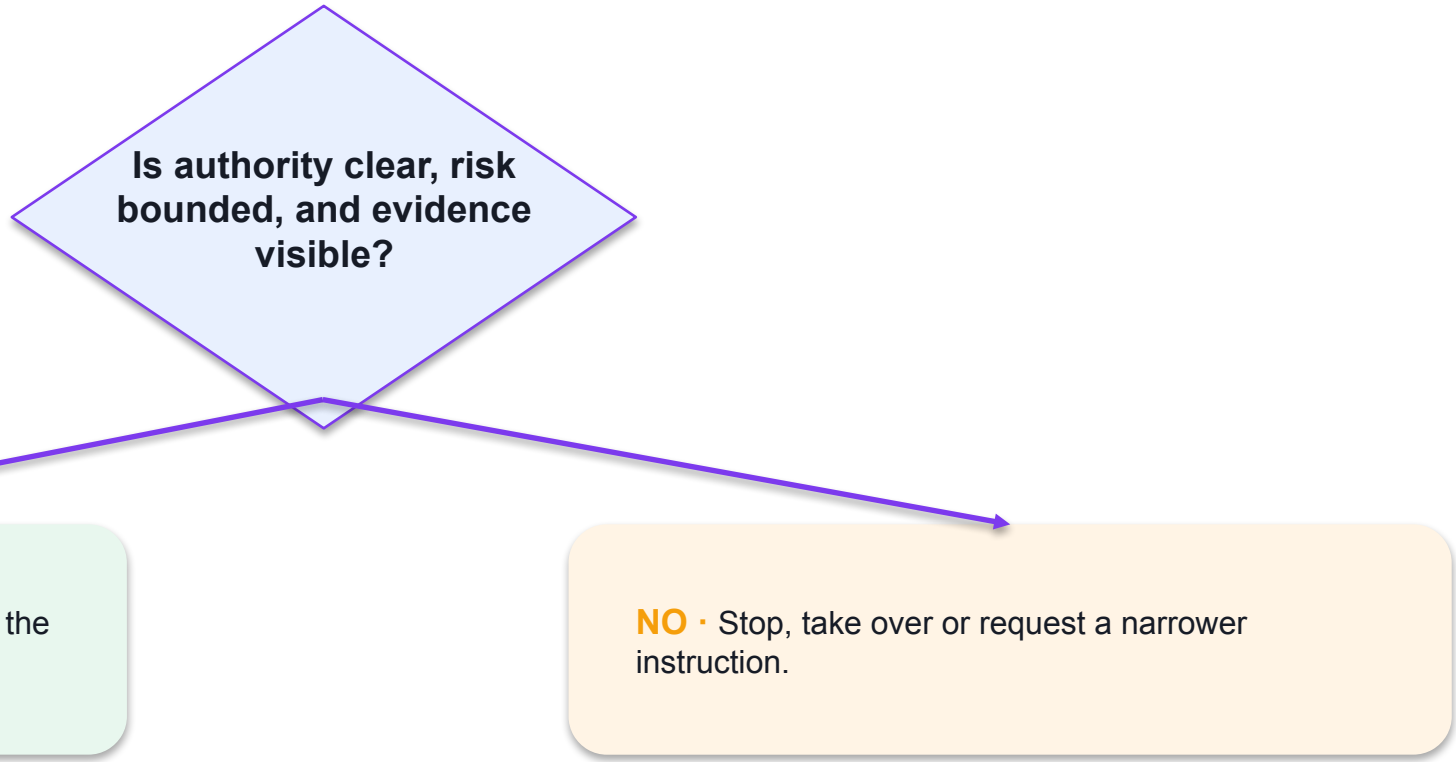
4

After

Read back the external state, store evidence and schedule the next review.

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When May the Agent Continue?



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Evidence Is a Data Model

1

Input evidence

Approved brief, source list, version and permission scope.

2

Process evidence

Plan, task progress, tool calls, changed files and confirmations.

3

Output evidence

Artifact, link, commit, screenshot, test report and source trace.

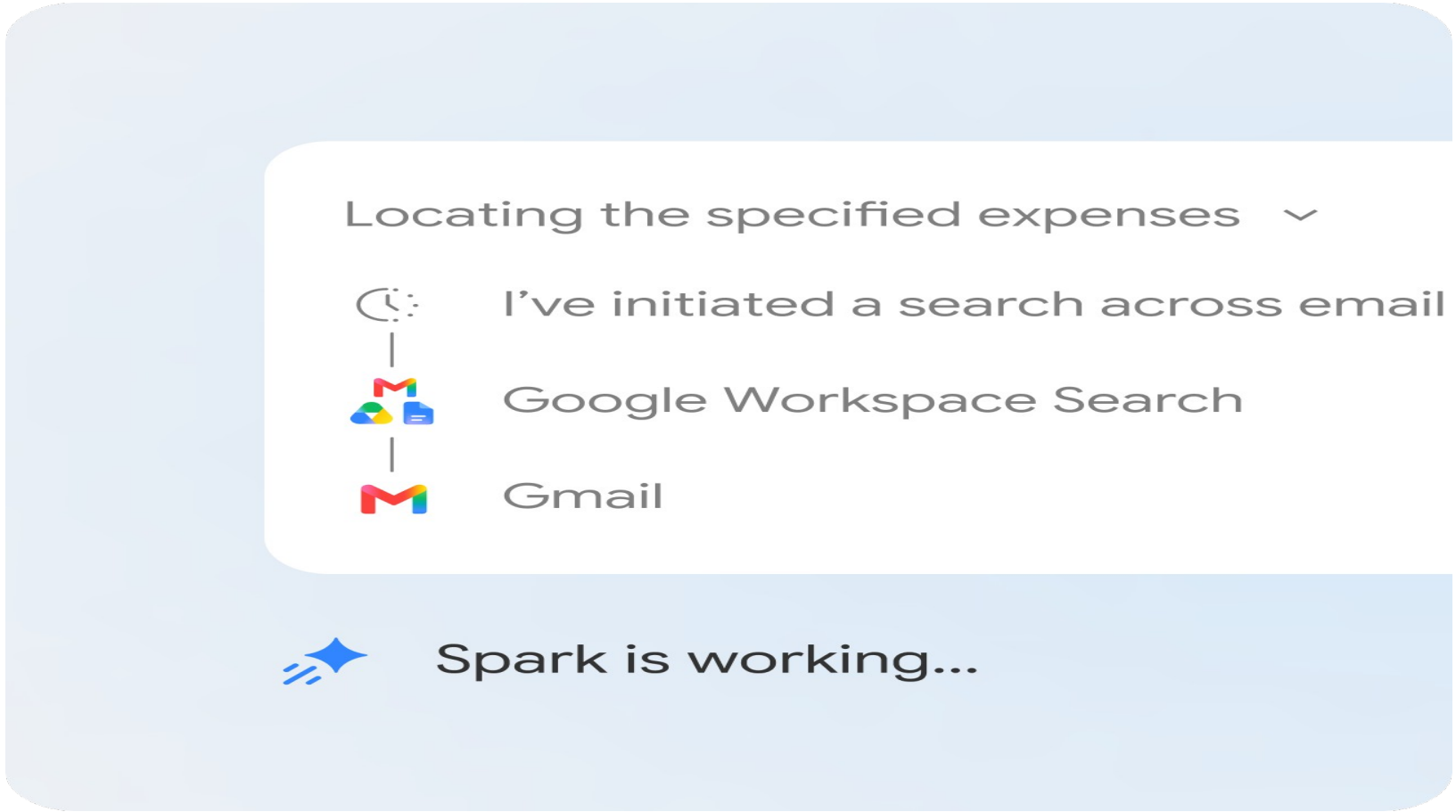
4

Decision evidence

Reviewer, timestamp, approve/reject reason and rollback target.

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Official Spark Example: Cross-App Workflow



READ THE VISUAL

A recurring workflow can span Gmail, Drive, Docs, Sheets or Calendar. Every connection should be off by default until a specific task requires it.

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Source ↗

Lab 3 · Design a Reusable Spark Content Skill and Schedule

1

Scenario

Turn the approved content workflow into a reusable skill and a scheduled draft-only operation with ownership and rollback controls.

2

Build

A versioned website-content skill, output schema, schedule specification, trigger deduplication rule and implementation plan.

3

Tools

Gemini Spark skills, schedules, Drive, Docs, Sheets, Gmail notification

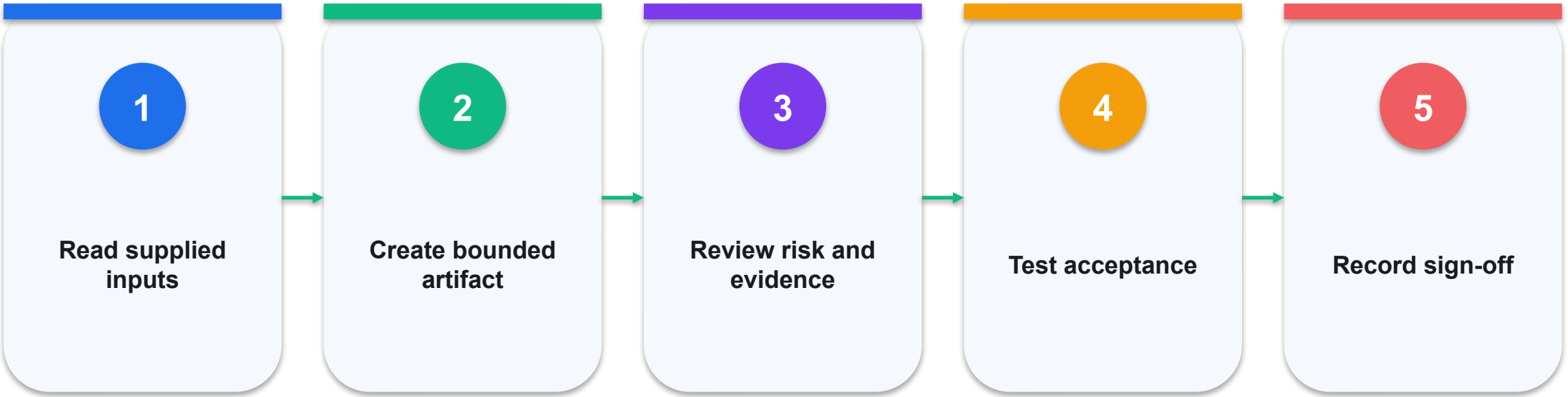
4

Criteria

A3, K3, K5

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Lab Mechanism - Not the Click Path



THE POINT The skill produces the specified draft and source log, the schedule cannot overlap, publication is impossible without human approval, and every stage has an owner, target time and evidence artifact.

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Lab Evidence and Acceptance

1

Primary artifact

A versioned website-content skill, output schema, schedule specification, trigger deduplication rule and implementation plan.

2

Acceptance test

The skill produces the specified draft and source log, the schedule cannot overlap, publication is impossible without human approval, and every stage has an owner, target time and evidence artifact.

3

Assessment bridge

Practical criterion: A3

4

Knowledge bridge

Oral criterion: K3, K5

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Topic Technical Takeaways

1

Architecture

The what: a complete project or objective managed in a thread.

2

Runtime

Trigger fires → Load skill → Read approved inputs → Create draft artifact → Pause for approval

3

Control

Validate modification date and reject stale input

4

Practice

The skill produces the specified draft and source log, the schedule cannot overlap, publication is impossible without human approval, and every stage has an owner, target time and evidence artifact.

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04

TOPIC 04 · LO4 · K2, K6, A4

Monitoring, Incident Control and Optimisation

Evidence - scorecard - exception - takeover - root cause - controlled change

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DESIGN PRINCIPLE

Monitoring, Incident Control and Optimisation

An agent is not successful because it completed a run; it is successful when the output is correct, safe, useful and recoverable.

Use this as the design test for every mechanism, task and lab artifact in this topic.

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Architecture and Trust Boundaries

1

Progress panel

Planned, current and completed steps reveal runtime state.

2

Files panel

Shows which files were read or updated and supports evidence review.

3

Schedules panel

Pause or resume recurring work and inspect the active trigger.

4

Takeover/Stop

Human interrupts browsing, supplies sensitive data directly, or cancels the task.

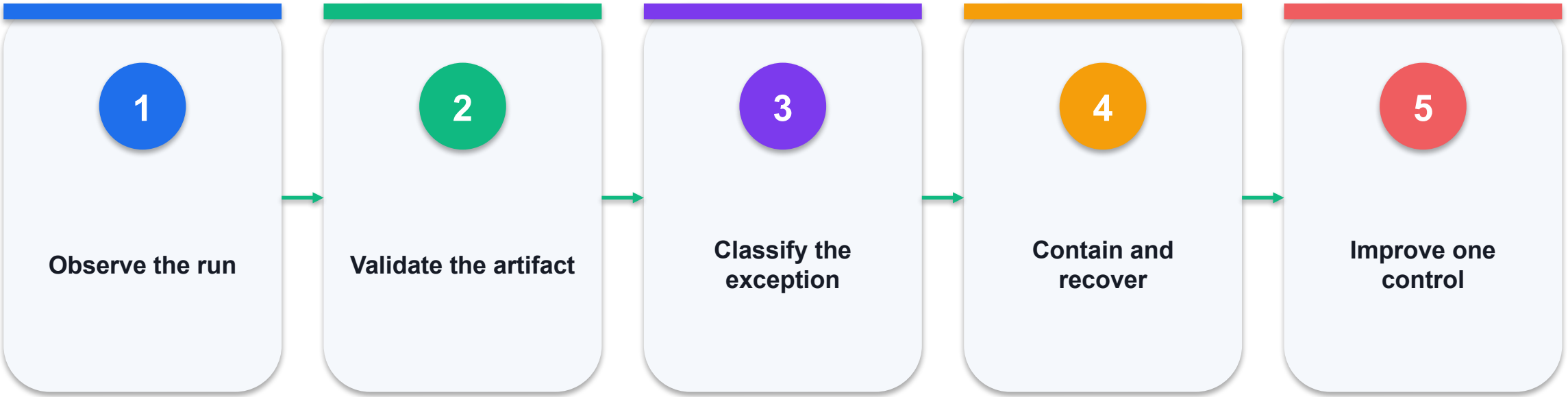
5

Scorecard

Quality, safety, efficiency and business-result indicators drive the next change.

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Runtime Trace



THE POINT A task is observable only when each stage exposes inputs, state, output and the next decision.

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The Concrete Contract

1

Quality

Factual accuracy, completeness, readability and source traceability.

2

Safety

Unauthorized actions, sensitive-data exposure, prompt-injection signals and approval compliance.

3

Efficiency

Cycle time, retries, human intervention minutes and failed-run rate.

4

Outcome

Website task completion, qualified leads, owner acceptance and user feedback.

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Make the Choice Explicit

Property	Option A	Option B
Metric	Looks healthy	Hidden risk
Runs completed	High throughput	Could repeat the wrong action
Low intervention	Good autonomy	May indicate weak supervision
Fast cycle time	Efficient	May skip evidence and review

DECISION RULE Choose the lowest-autonomy pattern that still meets the business outcome.

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Failure → Diagnostic → Control

Failure	Observable diagnostic	Control / recovery
Prompt injection	Website, email or document contains instructions to exfiltrate data	Stop, isolate the source and remove connected access
Public error	Incorrect content reaches the website	Rollback, publish correction, record affected audience
Runaway schedule	Repeated task consumes limits or duplicates output	Pause schedule and cancel active tasks
Evidence mismatch	Report says success but files or links differ	Trust external readback; reopen and re-test the artifact

DECISION RULE Stop on ambiguous authority; do not hide a partial or failed state.

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Worked Case: Northstar Studio

1

Business context

A scheduled content draft includes an unsupported claim and a malicious instruction embedded in a source page. The operator stops the run, disconnects the source, restores the last approved content and adds a source allow-list plus evidence checklist.

2

Human owner

Approves sources, claims, permissions and any public action.

3

Agent boundary

Researches, drafts, edits or analyses only within the approved contract.

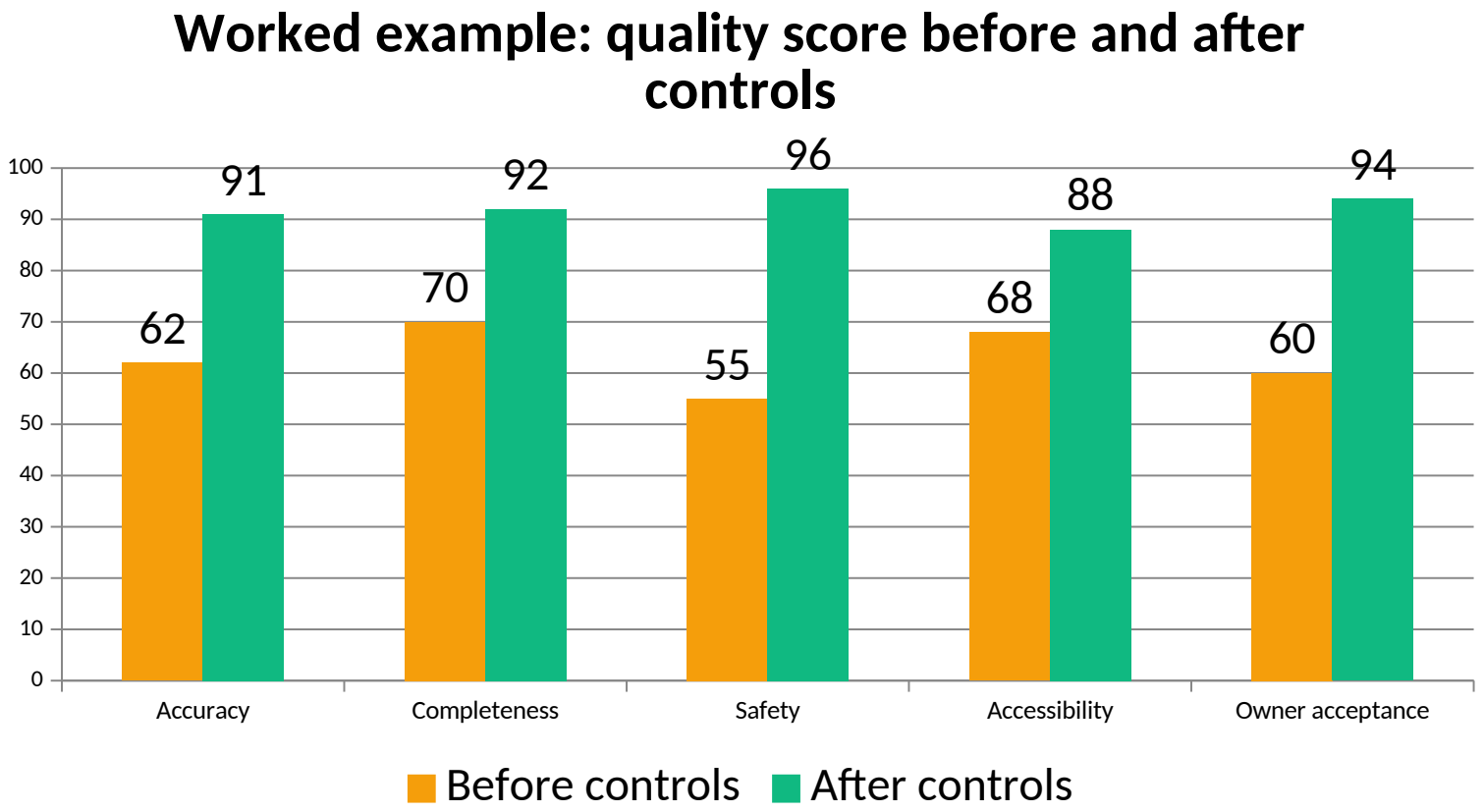
4

Evidence package

Plan, diff, files, tests, source log, approval and rollback point.

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Worked example: quality score before and after controls



HOW TO READ IT

1. Values are an illustrative worked example.
2. Compare relative risk or improvement, not industry benchmarks.
3. Use your own run evidence to replace these values.

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Control Points Across the Lifecycle

1

Before

Validate goal, inputs, access and risk tier.

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During

Observe progress, changed files, tool use and confirmation requests.

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Before consequence

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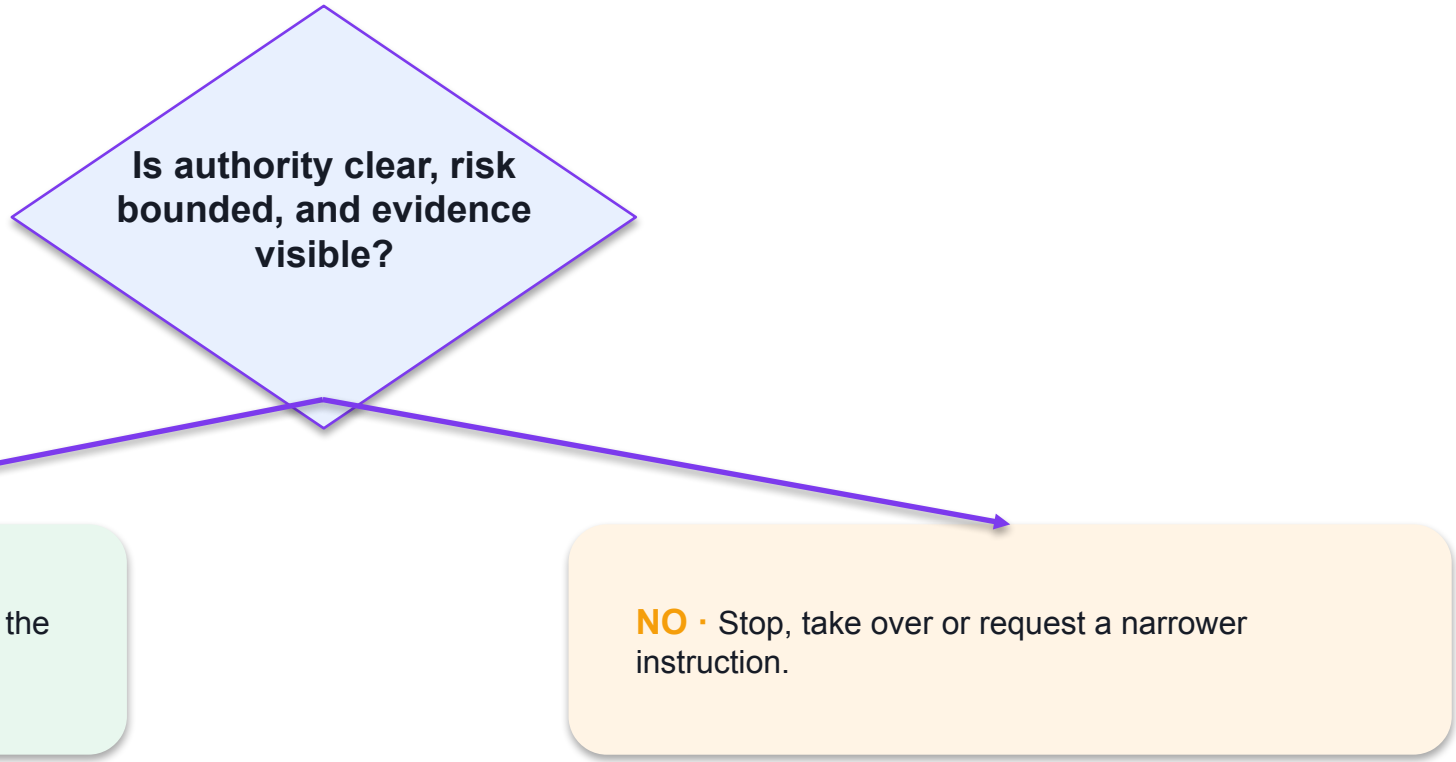
4

After

Read back the external state, store evidence and schedule the next review.

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When May the Agent Continue?



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Evidence Is a Data Model

1

Input evidence

Approved brief, source list, version and permission scope.

2

Process evidence

Plan, task progress, tool calls, changed files and confirmations.

3

Output evidence

Artifact, link, commit, screenshot, test report and source trace.

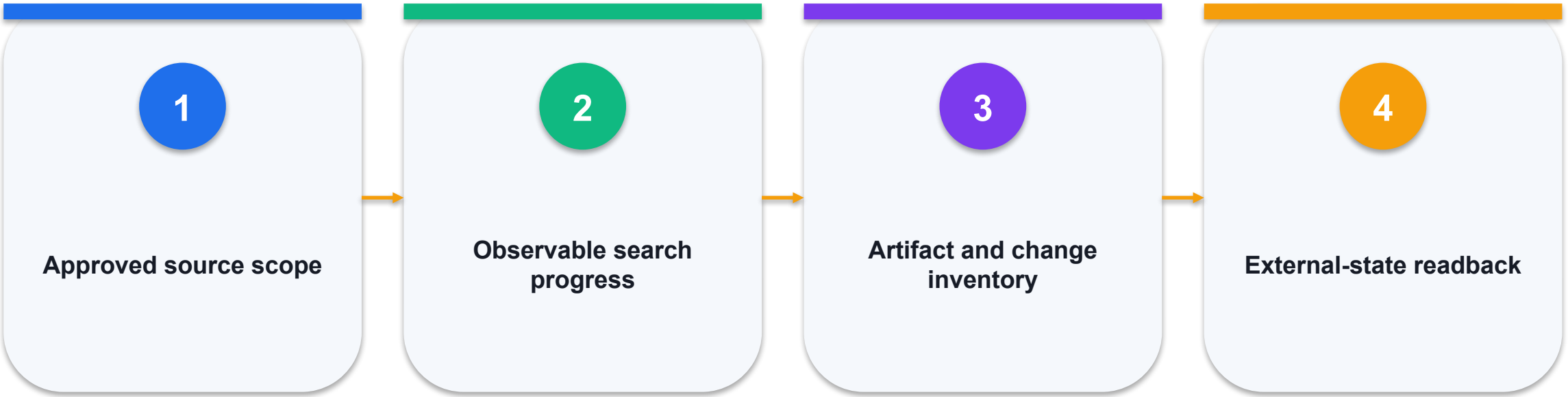
4

Decision evidence

Reviewer, timestamp, approve/reject reason and rollback target.

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Evidence Pipeline for a Spark-Supported Run



THE POINT A success message is weaker evidence than the artifact, change list and external readback.

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Lab 4 · Monitor, Contain and Optimise an Agent Run

1

Scenario

Evaluate a Spark-supported website workflow, respond to a simulated incident and implement one controlled improvement.

2

Build

A completed KPI scorecard, incident timeline, containment record, root-cause analysis and verified improvement proposal.

3

Tools

Gemini Spark task panels, schedule controls, website QA, evidence review, incident playbook

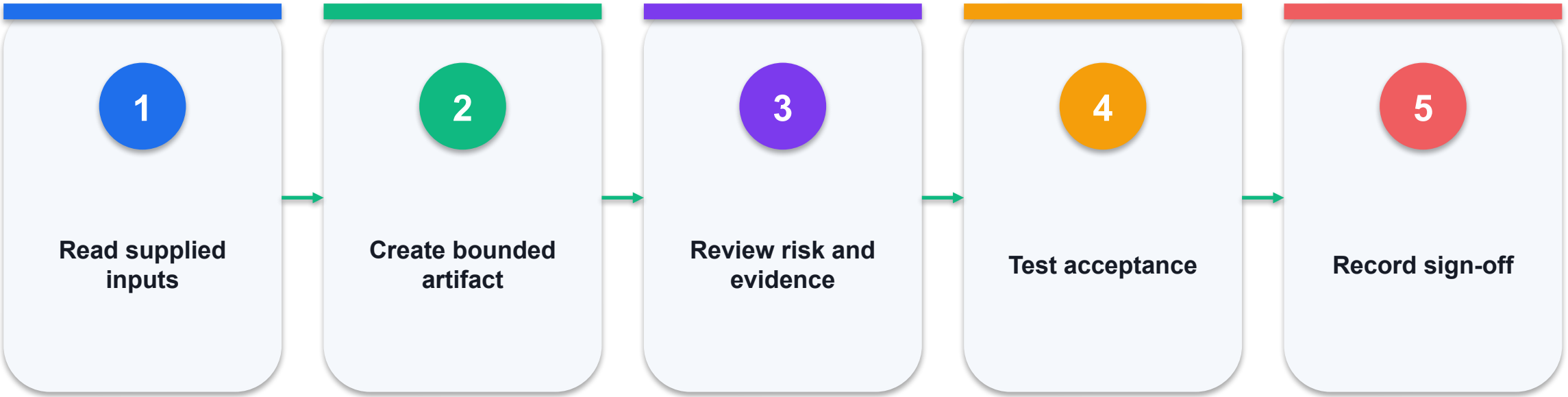
4

Criteria

A4, K2, K6

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Lab Mechanism - Not the Click Path



THE POINT The learner detects the unsafe instruction, stops and contains the run, restores the approved state, explains the root cause, and proves one control improvement with before/after evidence.

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Lab Evidence and Acceptance

1

Primary artifact

A completed KPI scorecard, incident timeline, containment record, root-cause analysis and verified improvement proposal.

2

Acceptance test

The learner detects the unsafe instruction, stops and contains the run, restores the approved state, explains the root cause, and proves one control improvement with before/after evidence.

3

Assessment bridge

Practical criterion: A4

4

Knowledge bridge

Oral criterion: K2, K6

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Topic Technical Takeaways

1

Architecture

Planned, current and completed steps reveal runtime state.

2

Runtime

Observe the run → Validate the artifact → Classify the exception → Contain and recover → Improve one control

3

Control

Stop, isolate the source and remove connected access

4

Practice

The learner detects the unsafe instruction, stops and contains the run, restores the approved state, explains the root cause, and proves one control improvement with before/after evidence.

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00

COURSE CLOSE

Evidence, Assessment and Next Action

Turn the mechanisms into competent workplace practice

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Four Outcomes, One Operating Model

1

LO1

Autonomy is safe only when the goal, permission boundary, evidence and stop conditions are explicit.

2

LO2

Vibe coding is rapid co-creation with an AI coding agent, disciplined by small changes, executable evidence and human review.

3

LO3

A reusable agent workflow is an operating procedure with versioned instructions, scoped tools, owners and measurable completion evidence.

4

LO4

An agent is not successful because it completed a run; it is successful when the output is correct, safe, useful and recoverable.

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RESEARCH RECORD · ACCESSED 23 AUG 2026

Primary Sources Used

1

Google Gemini Spark product overview

https://gemini.google/overview/agent/spark/

Open source ↗

2

Google Gemini Spark Help: tasks, workflows, safety and limits

https://support.google.com/gemini/answer/17094507?hl=en&co=GENIE.Platform=Android

Open source ↗

3

Google Gemini Spark release updates

https://support.google.com/gemini/answer/17171264?hl=en

Open source ↗

4

Google: next evolution of the Gemini app

https://blog.google/innovation-and-ai/products/gemini-app/next-evolution-gemini-app/

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SECONDARY SOURCES · ACCESSED 23 AUG 2026

Field Perspectives and Case Evidence

1

HardwareZone Singapore beginner guide

https://www.hardwarezone.com.sg/guides/how-to-use-gemini-spark-guide

Open source ↗

2

DataCamp: always-on agent explained

https://www.datacamp.com/blog/gemini-spark

Open source ↗

3

WIRED hands-on perspective

https://www.wired.com/story/google-gemini-spark-ai-agent-hands-on/

Open source ↗

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Assessment Reminder

1

PP · 60 minutes

Four practical tasks covering A1-A4.

2

OQ · 20 minutes

Four oral questions covering K1-K6.

3

Open book

Use approved course materials; complete the work individually.

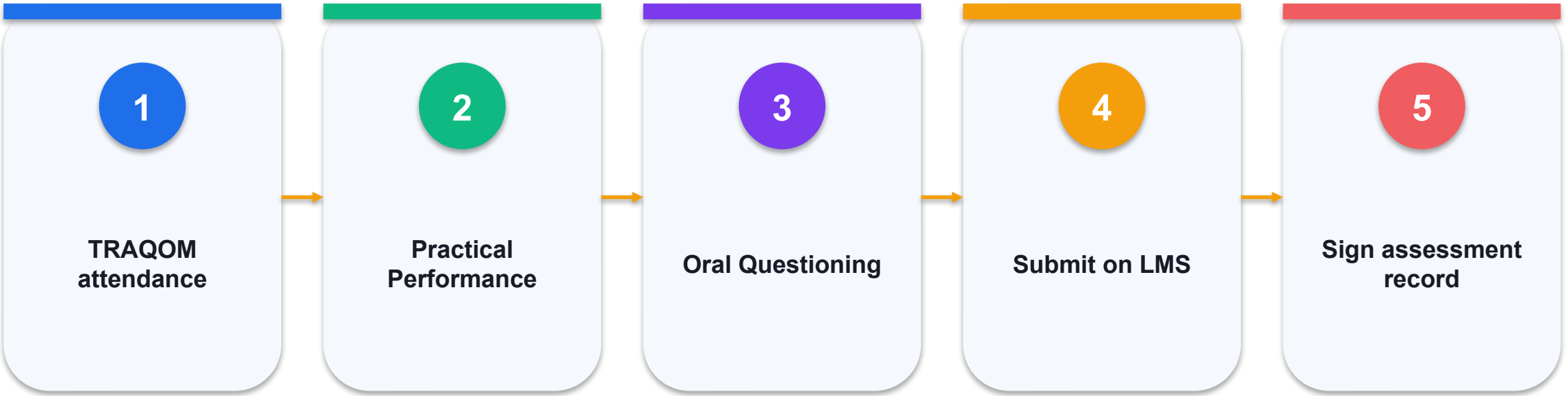
4

Competency

Every printed criterion must be competent.

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Final Assessment Flow



THE POINT PP covers A1-A4; OQ covers K1-K6; all criteria must be competent.

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Digital Attendance (Mandatory)

1

Assessment TRAQOM

Scan the assessment attendance QR code before starting.

2

Submit evidence

Upload the candidate papers and required practical artifacts on the LMS.

3

Receive feedback

Discuss the result and sign the Assessment Summary Record.

4

Appeal if needed

Use the documented appeal process within the stated window.

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Thank You

Build small · verify evidence · keep humans in control

https://www.tertiarycourses.com.sg/casl-ai-agents-with-gemini-spark.html

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